

Weekly Research Update

Three Project Progress Report

Muting (Don) Ma

September 2, 2026

Embedding Retrieval Capacity

Bi-objective Trajectory Revision C1

Stochastic Multiple Knapsack Manuscript

Public artifacts: [OpenProjects repository](#)

This report summarizes research progress since the August 27, 2026 meeting. Each project page defines its setting, separates completed evidence from open work, identifies decisions requested from Dr. Gong, and links to a public snapshot. Primary references follow the three project pages.

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1. Embedding Retrieval Capacity

Name: Muting (Don) Ma

Project: Embedding Retrieval Capacity for Decision Support

Date: September 2, 2026

Context. Dense retrieval encodes each query and document as one fixed-length vector, then ranks by geometric similarity. This project studies when that interface loses relevance patterns and whether the resulting errors damage optimization decisions.

1. Current objective

- Relate retrieval-capacity limits to feasibility loss and operational regret in a proposal-ready guarantee.

2. Progress since the last meeting

- **Completed:** Built a 72-slide plain-language dissection of Weller et al.'s theorem, free-vector experiment, LIMIT benchmark, and appendices.
- **Completed:** Separated proved capacity from empirical observations and design implications to avoid overstating evidence.
- **In progress:** Adapted the paper's family-specific bound to the NSF Operations Engineering proposal.

3. Key result or lesson

- For n documents, returned-set size k , robustness margin 2γ , and vector dimension d , realizing every size- k answer requires $d \geq \log \binom{n}{k} / \log(1 + 1/\gamma)$ [1].
- LIMIT uses simple queries with overlapping relevant pairs and shows trained models use dimensions less effectively than optimized free vectors.
- The theorem is a capacity floor, not a recall prediction, so the decision-level consequence remains the open proposal contribution.

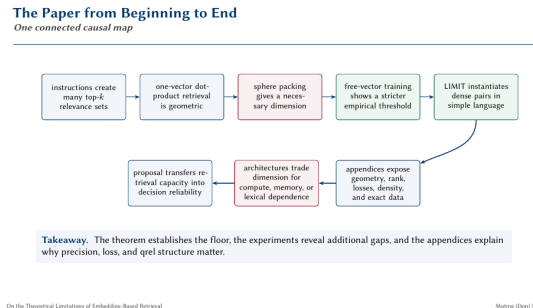


Figure 1: Logic connecting fixed-length retrieval capacity to decision reliability.

4. Input needed from Dr. Gong

- Should the transfer theorem target feasibility loss, objective regret, or both?
- Should the first stress test compare the single-vector baseline with a multi-vector method, a lexical-neural hybrid, or both?

5. Before the next meeting

- Draft the proposal subsection.
- Formalize the first error-to-regret proposition.
- Specify a toy operational stress test and metrics.

Public artifacts: [project folder](#); [selected snapshot](#).

2. Bi-objective Trajectory Revision C1

Name: Muting (Don) Ma

Project: Bi-objective Intersection Trajectory Revision C1

Date: September 2, 2026

Context. At an automated intersection, the paper trades vehicle delay against modeled crossing collision cost. Revision C1 corrects how repeated conflict-cell encounters for a vehicle pair become one probability and loss.

1. Current objective

- Close revision C1 with auditable pair risk, calibrated scale, practical controller comparisons, and reproducible evidence.

2. Progress since the last meeting

- **Completed:** The revision/c1 branch now uses one arrival-time error draw and the smallest shared-cell entry gap per vehicle pair.
- **Completed:** Regenerated costs, the control menu, break-even results, figures, and claims after removing the retired average and count detour.
- **In progress:** The main audit finds only Step 1 of nine complete, so the revision branch remains open.

3. Key result or lesson

- For vehicles q and q' , $\gamma_{\min}$ is the smallest entry-time gap, p its event probability, and κ_c the collision price.
- The schedule cost is $C^{\text{cross}} = \kappa_c \sum_{\{q, q'\}} p(\gamma_{\min}(q, q'))$, and linearity permits the sum without pair independence.
- The eight-vehicle case still favors coordination over the fallback but not extra speed; absolute costs require calibration.

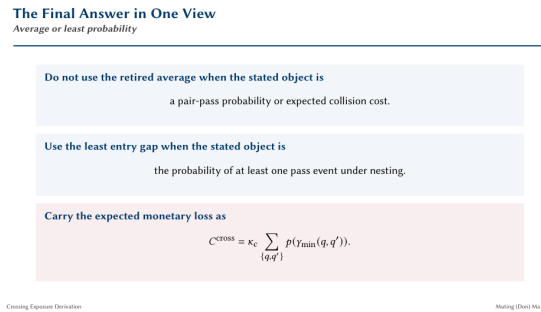


Figure 2: Least-gap construction that converts pair interactions into schedule expected cost.

4. Input needed from Dr. Gong

- Should the one-draw least-gap construction remain the primary risk model while calibration is developed?
- Should the next work prioritize the audit and calibration package or practical-controller pilots?

5. Before the next meeting

- Export canonical pair records and the pair-level audit.
- Run the prescribed error-scale sensitivity.
- Specify controller comparisons and seed and demand pilots.

Public artifacts: project folder; selected snapshot.

3. Stochastic Multiple Knapsack Manuscript

Name: Muting (Don) Ma

Project: Stochastic Multiple Knapsack Manuscript

Date: September 2, 2026

Context. The manuscript assigns uncertain tasks in a two-stage multiple-knapsack model. Its Benders method separates initial assignments from scenario subproblems, and the review checks the human-centered claim, integer decisions, and experiments against the code.

1. Current objective

- Rebuild the paper around a defensible contribution, feasible integer assignments, and executable stochastic, human-outcome, and algorithmic benchmarks.

2. Progress since the last meeting

- **Completed:** Produced a full dissection of the formulation, decomposition, results, ten advisor points, and revision order.
- **Completed:** Reviewed two scheduling studies for fairness and uncertainty evidence and two knapsack studies for algorithm evidence [2, 3, 4, 5].
- **Blocked:** The claim remains blocked because the feasibility proposition is false and reported assignments come from a continuous relaxation.

3. Key result or lesson

- The defensible core is a time-discounted formulation plus a decomposition proof of concept for its continuous relaxation.
- An all-zero solution refutes the feasibility claim, and 12 of 21 assignment variables are fractional rather than feasible integer decisions.
- The closest literature turns the broad novelty claim into three tests: stochastic value, human-outcome value, and algorithmic value.



Figure 3: Manuscript strengths, publication blockers, evidence obligations, and revision verdict.

4. Input needed from Dr. Gong

- Should the paper narrow to managerial task scheduling or add an explicit individual outcome and fairness treatment?
- Should the first integer baseline use direct mixed-integer optimization, integer Benders, or rounding and repair?

5. Before the next meeting

- Correct the proposition and prove nontriviality.
- Produce feasible integer incumbents and gaps.
- Prespecify held-out tests, seeds, scaling, and ablations.

Public artifacts: [project folder](#); [selected snapshot](#).

References

- [1] O. Weller, M. Boratko, I. Naim, and J. Lee, “On the Theoretical Limitations of Embedding-Based Retrieval,” International Conference on Learning Representations, 2026. [ICLR paper page](#).
- [2] E. Gülmez, M. E. Aydin, K. B. Urganci, S. C. Gamoura, and H. I. Koruca, “Healthcare Staff Scheduling with Work-Life Balance Constraint Using Multi Objective Evolutionary Algorithms,” *Computers and Industrial Engineering*, vol. 220, 2026. doi:10.1016/j.cie.2026.112287.
- [3] C. Perreault-Lafleur, M. Carvalho, and G. Desaulniers, “A Stochastic Integer Programming Approach to Reserve Staff Scheduling with Preferences,” *International Transactions in Operational Research*, vol. 32, no. 1, pp. 289 to 313, 2025. doi:10.1111/itor.13298.
- [4] S. Mancini, C. Meloni, and M. Ciavotta, “A Decomposition Approach for Multidimensional Knapsacks with Family-Split Penalties,” *International Transactions in Operational Research*, vol. 31, no. 4, pp. 2247 to 2271, 2024. doi:10.1111/itor.13207.
- [5] M. Masmoudi, Y. Adouani, and B. Jarboui, “LP Relaxation and Dynamic Programming Enhancing VNS for the Multiple Knapsack Problem with Setup,” *International Transactions in Operational Research*, vol. 31, no. 3, pp. 1890 to 1916, 2024. doi:10.1111/itor.13213.